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Visualizing Mean, Median, Mean Deviation and Standard Deviation of a Set of Numbers

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Abstract

We review the existing visualizations of the mean and the median of a given set of numbers. Then we give an alternative visualization of the mean using the empirical cumulative distribution function of the given numbers. Next, we visualize the mean deviation (MD) and the mean square deviation (MSD) of the given numbers from any arbitrary value, including the variance. In light of these new visualizations, we revisit the well-known optimal properties of the MD from the median and the MSD from the mean. We also give a more elementary explanation of why the denominator of the sample variance of a set of numbers is one less than the sample size.

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1. Introduction

Deviation is a basic concept in probability and statistics. We shall use the word "deviation" to mean "absolute difference." Thus, the deviation between two numbers a and b is $|a - b|$. Likewise, the deviation of each number a and b from their mid-point or their average $(a + b)/2$, is half the absolute difference between them, or $|a - b|/2$. Given $n > 2$ numbers, we seek to find a single measure of spread that combines all possible pairwise deviations in a suitable way. The most widespread notion of this combined deviation is the standard deviation (SD); and a distant second is the mean (absolute) deviation (MD). What exactly are these quantities? Why is the former so popular relative to the latter and among many other notions of deviations?

To answer these questions, we must first address another basic question: What quantity do we wish to measure the deviations from? Such a quantity is often a measure of center. Again, there are several choices for a measure of center—the most common ones are the mean (or arithmetic mean) and the median. Other choices for the measure of center (such as the mode or the geometric mean or the harmonic mean) may be more appropriate in certain special situations. But we won't consider them in this paper. We emphasize that the interpretation of any particular choice of spread goes hand in hand with the corresponding choice of center from which such a spread is measured.

We give the geometric interpretations of the common measures of center in Section 2, and those of spread in Section 3. Section 4 concludes the paper with some discussions.

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2 Measures of Center

2.1.1 The Mean as a Fulcrum

The mean is the most frequently used measure of center and is the most basic concept in statistics and experimental sciences. It occurs frequently in many everyday life applications (Pollatsek, Lima and Well, 1981; Lesser, Wagler and Abormegah, 2014). Suppose that we have *n* numbers $x_1, x_2, ..., x_n$. The mean is given by

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad (1)$$

The geometric interpretation of the mean as a fulcrum is very well-known. See, for example, Pollatsek, Lima and Well (1981); Mokros and Russell (1995); Embse and Engebretsen (1996); Doane and Tracy (2000). We illustrate it in Examples 1 and 2.

Example 1 Suppose that we are given five numbers: 1, 4, 7, 10, and 13. The mean is 35/5=7. So, if we put the fulcrum at 7, the system remains perfectly balanced.

Example 2 Suppose that we have five data points as in Example 1, except that the largest value 13 is changed to 23. See Fig. 2. The fulcrum should move to the new mean 45/5=9.

Visualizing the mean as a fulcrum helps explain that the mean is sensitive to a change in any one (or more) of the numbers. However, finding the location of the fulcrum, without doing the actual computation for the mean, is prone to occasional mistakes. In the next Subsection we give an alternative interpretation of the mean, which will reduce the pitfall of guessing the fulcrum.

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2.1.2 The Mean as an Area and as the Location of a Vertical Line

Let us, without loss of generality, assume that all n given numbers are **positive**.[1] Draw the empirical cumulative distribution function (ECDF) of these data. See Fig. 3(a) (which uses the data in Example 1).

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Recall that the ECDF is a step function which has jumps of height $1/n$ occurring at each of the given numbers. If we extend the step levels all the way to the left (shown by dotted lines) until they touch the y-axis, then the given numbers are represented by the widths of the rectangles so formed (with each rectangle having height $1/n$). Also these rectangles are stacked without any gap or overlap. Therefore, the total area of the shaded region in Fig. 3(b) is exactly $\bar{x}$.

Next, we exhibit in Fig. 3(c) a rectangle $OPQR$ (of height 1) whose area equals that of the shaded region in Fig. 3(b). To obtain such a rectangle, one simply needs to locate the right edge PQ appropriately. Of course, the width OP of this rectangle (of height 1) also equals $\bar{x}$. Superimposing Fig. 3(b) on Fig. 3(c), and un-shading the portions of the shaded regions that overlap with each other, we establish that the two remaining shaded regions in Fig. 3(d) have equal areas. Thus, PQ is a vertical line ℓ that intersects the graph of the ECDF is such a way that the areas of the shaded regions in Fig. 3(d) are equal. This gives us a new geometric way to visualize the mean as a vertical line that equalizes the two shaded areas in Fig. 3(d).

[1] If not, add a sufficiently large constant C to all numbers to make them all positive. Then consider the mean of the transformed values. Finally, subtract C from that calculated mean to recover the mean of the given values.

Thus, to visualize the mean, on the graph of the ECDF we contemplate a vertical line ℓ that is capable of sliding from left to right. See Fig. 4 (a). Where must we stop ℓ in order to ensure that $A_\ell = B_\ell$? Here, A_ℓ is the area of the region to the right of the vertical line ℓ, above the ECDF, and below the horizontal line at height 1; and B_ℓ is the area of the region to the left of the line ℓ, below the ECDF, and above the *x*-axis. Note that as we slide the vertical line ℓ from left to right, A_ℓ decreases and B_ℓ increases continuously (with a positive derivative). Therefore, by Rolle's Theorem (see, Anton et al. 2012, p. 302), there is a unique ℓ such that $A_\ell = B_\ell$. In Fig. 3(d) and Fig. 4(b) we have stopped ℓ at $\ell = \bar{x}$ in order to ensure that $A_\ell = B_\ell$.

Indeed, something more is true. If we locate the vertical line at *any* arbitrary location ℓ, as in Fig. 4(a), we can already visualize the mean $\bar{x}$ as ℓ plus the difference $A_\ell - B_\ell$, simply by comparing Fig. 3(b) and Fig. 4(a). Therefore, we have

Proposition 1 *For a given set of n numbers, the mean can be found as*

$$\bar{x} = \ell + (A_\ell - B_\ell) \quad \textit{for all} \quad \ell \in \mathfrak{R} \tag{2}$$

In particular, if ℓ is chosen to be $\ell = \bar{x}$, we will have $A_{\bar{x}} = B_{\bar{x}}$.

2.2.1 The Median as the Middle-most Value

Another widely used measure of center is the median, which is the middle-most value of an *ordered* set of numbers. If the values are $x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$, then the median *m* is given by

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$$m = \begin{cases} x_{(k+1)} & \text{if } n = 2k+1 \\ \left(x_{(k)} + x_{(k+1)}\right)/2 & \text{if } n = 2k \end{cases} \quad (3)$$

In both Examples 1 and 2, the median is 7, even though the largest values differ. We say the median is a *robust* statistic; that is, it is not sensitive to a change in the extreme value(s). The next example illustrates the median in case of an even number of given values.

Example 3 Suppose that we are given six numbers: 10, 7, 4, 19, 1, and 13. The median is between the third and the fourth numbers in the ordered list: 1, 4, 7, 10, 13, 19. So, any number in [7, 10] is a median, including the mean 9. We take the mid-value 8.5 of [7, 10] to be the (automated) median. Fig. 5 shows both the mean and the median with reference to the dot plot.

2.2.2 The Median Obtained from the ECDF

The geometric interpretation of the median using the ECDF is very well-known. See Fig. 6.

On the graph of the ECDF of the data, draw the horizontal line y=½, intersecting the ECDF either at a point P=(m, ½) as in Fig. 6(a), or in an interval $I = [m_L, m_U] \times \{1/2\}$ as in Fig. 6(b). In Fig. 6(a), the abscissa of P is the unique median at m=7. In Fig. 6(b), any value in the interval $[m_L, m_U]$ is a median. By convention, we take the midpoint of the interval as the median.

3 Measures of Spread

How far are the given values from any specified number ℓ (such as a median, or the mean, or 0, or any other constant)? No matter which ℓ we choose, it is easy to calculate the difference $x_i - \ell$,

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or the deviation $|x_i - \ell|$ for *each* given value x_i, $i = 1, 2, ..., n$. But how does one combine the *n* individual differences or deviations into one single measure of spread? There are several favorite methods to combine them. Accordingly, we get different measures of spread.

3.1.1 Mean Deviation as an Average

Since $\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\,\bar{x} = 0$, the mean of the differences of the given values from the mean of all the values is *always* 0. What if we consider the deviations (or the absolute differences) of the given values from the mean? Then the mean of these *n* deviations is called the mean deviation from the mean $\bar{x}$, or

$$MD(\bar{x}) = \frac{1}{n}\sum_{i=1}^{n} |x_i - \bar{x}| \tag{3}$$

Likewise, we can define the mean deviation of the given values from any number ℓ as

$$MD(\ell) = \frac{1}{n}\sum_{i=1}^{n} |x_i - \ell| \tag{4}$$

Thus, $MD(\ell)$ is an average; that is, $MD(\ell) = \overline{w}_\ell$, where $w_\ell = |x - \ell|$. In particular, the MD from the median *m* is $MD(m) = (\sum |x_i - m|)/n$; the MD from 0 is $MD(0) = (\sum |x_i|)/n$; and whenever all given values are nonnegative (as in Examples 1 through 3), $MD(0) = \bar{x}$. The computations of mean deviations in Examples 1 through 3, are summarized below.

In all Examples 1–3, $MD(m) \le MD(0)$ and also $MD(m) \le MD(\bar{x})$. In Example 3, even though any number in [7, 10] is a median, MD(*m*) is a constant for all medians *m* in [7,10], including the mean $\bar{x} = 9$. These illustrate the well-known universal result, stated in Proposition 2 below.

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Proposition 2 *$MD(\ell)$ is minimized when deviations are measured from any median m.*

For a typical proof of Proposition 2, the reader may see Dwight (1957). However, to visualize Proposition 2, it is instructive to draw $MD(\ell)$ as a function of ℓ. For each of the given numbers x_i, note that $|x_i - \ell|$ is a linear function of ℓ with slope -1 for $\ell < x_i$ and 1 for $\ell > x_i$. Being the average of these linear functions, $MD(\ell)$ is a piecewise linear function of ℓ, with slopes of the linear pieces from left to right in arithmetic progression ranging from -1 to 1. Fig. 7(a) and 7(b) illustrate this feature in Examples 1 and 3. Clearly, $MD(\ell)$ is minimized at the unique median *m* or at *any* median (when there is an interval of medians). We will give an alternative proof of Proposition 2 after giving an interpretation of $MD(\ell)$ in the next Subsection.

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3.1.2 Mean Deviation as an Area and as a Width

If we superimpose a vertical line at ℓ on the graph of the ECDF, as in Fig. 4(a), we can visualize $MD(\ell)$ as the total area of the two shaded regions $A_\ell + B_\ell$. This follows from (4) by partitioning the shaded regions in Fig. 4(a) into rectangles of width $|x_i - \ell|$ and height 1/*n*. See Fig. 8 (a).

Next, to visualize $MD(\ell)$ as a distance, we construct a rectangle of height 1 whose left side coincides with ℓ and whose area equals $A_\ell + B_\ell$. Then $MD(\ell)$ is both the area and the width of this rectangle of height 1. See Fig. 8(b).

The geometric interpretation of $MD(\ell)$ as an area affords us an alternative proof of Proposition 2 as follows: Using the data in Example 1 as illustration, Fig. 9 shows the ECDF, denoted by $F_n(x)$, intersected by a vertical line at ℓ. As the vertical line moves from an arbitrary location $\ell < m$ to the median *m*, since $1-F_n(\ell) > F_n(\ell),$ the reduction in area A_ℓ of the checkered region in Fig. 9(a), exceeds the gain in area B_ℓ of the checkered region in Fig 9(b). Hence, $MD(\ell) = A_\ell + B_\ell$ decreases. Likewise, as the vertical line moves from an arbitrary location $\ell > m$ to the median *m*, since $F_n(\ell) > 1-F_n(\ell),$ the reduction in area B_ℓ exceeds the gain in area A_ℓ, causing $MD(\ell) = A_\ell + B_\ell$ to decrease.

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3.2.1 The Sample Standard Deviation Defined Algebraically

The sample variance s^2 and the sample standard deviation *s* of a set of *n* given numbers are introduced as early as middle school (see [9]) through their algebraic expressions given below:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right] \tag{5}$$

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = \sqrt{\frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right]} \tag{6}$$

We found some visual representations of the SD in the literature. See Maverick (1932) and Embse and Engebretsen (1996). However, these are only depictions of the SD *after* it has been calculated numerically, not geometric interpretations. Students are told that the SD is a "typical distance" of the numbers from the mean. But if "typical distance" is interpreted as "average

distance," we get $MD(\bar{x})$, not SD. The correct interpretation of the sample SD is the "root *mean* square deviation (R*M*SD) from the mean," where the first *mean* involves a division by (*n*-1).

Some observant students may ask why in defining the variance, we must divide by $(n-1)$ in (5), and not by *n*. One explanation is that we do it so that the sample variance will be an unbiased estimator of the population variance.[2] But oftentimes students have not yet learned the notion of estimation or unbiasedness. Another explanation (see, for example, Martin 2003) involves the notion of "degree(s) of freedom," since only $(n-1)$ differences are 'free to vary.'

Neither of the above explanations addresses the true meaning of the SD. Our interpretation of the SD, given in Subsection 3.2.4, makes it absolutely clear (albeit at the cost of some algebraic manipulations) why the denominator in the definition of sample variance in (5) is $(n-1)$, without invoking the notions of estimation, unbiasedness or degrees of freedom. But first, let us see what would happen if in (5) and (6) we used *n* as the divisor instead of $(n-1)$.

3.2.2 The Population Standard Deviation Defined Algebraically

If the denominators in (5) and (6) are modified to *n*, then the quantity so defined is called the **population variance**, denoted by σ^2, and its square root is called the **population standard deviation**, denoted by σ. That is,

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}[\sum_{i=1}^{n} {x_i}^2 \; - \; n\,\bar{x}^2] \tag{7}$$

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\,]} = \sqrt{\frac{1}{n}[\sum_{i=1}^{n} {x_i}^2 \; - \; n\,\bar{x}^2]} \tag{8}$$

[2]Even though s^2 is an unbiased estimator of σ^2, it is well known that *s* is **not** an unbiased estimator of σ, although it is less biased than would be if the denominator were *n*.

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Thus, students who had puzzled over why the denominator of the sample variance is (n–1), are now vindicated. Indeed, the population variance is the mean (involving a division by n) squared deviation (MSD) from the mean; and the population SD is the RMSD from the mean,

More generally, the MSD from any value ℓ is defined as

$$MSD(\ell) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \ell)^2 = \sigma^2 + (\ell - \bar{x})^2 \qquad (9)$$

Here the second equality follows from simple algebra. From (9) the following well-known optimality property of $MSD(\ell)$ follows immediately.

Proposition 3 $MSD(\ell)$ is minimized at $\ell = \bar{x}$, and $min_{\ell} MSD(\ell) = MSD(\bar{x}) = \sigma^2$.

Using (9), one can draw $MSD(\ell)$ as a (quadratic) function of ℓ in order to visualize the optimal property stated in Proposition 3. We leave it to the reader to do so.

3.2.3 The Population Standard Deviation as the Radius of a Cylinder

Suppose that we treat the given set of n values as a population in its own right. We shall interpret the population SD given in (8). In fact, we shall give a geometric interpretation to the RMSD from *any* arbitrary value $x = \ell$. Suppose that the strip of plane between horizontal lines y=0 and y=1, containing the graph of the ECDF, is revolved about the vertical line $x = \ell$. Then the graph of the ECDF generates a collection C_ℓ of right circular cylinders with a common axis $x = \ell$, all with the same height 1/n, but with different cross-sectional radii $|x_i - \ell|$, as shown in Fig 10(a) left panel. Next, we consider a new right circular cylinder of **constant** radius (to be determined) and height 1, whose volume equals that of the total volume of the collection C_ℓ. When we have

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found such a cylinder, shown in Fig. 10(a) right panel, the constant radius of this new cylinder equals RMSD(ℓ).

Of course, we could choose $x = \bar{x}$ as the axis of revolution, in which case the corresponding RMSD($\bar{x}$), as described above, is shown in Fig. 10(b). Proposition 3 stated the well-known result that RMSD(x) is minimum when $x = \bar{x}$. This special RMSD($\bar{x}$) is the population SD σ.

Instead of the collection $C_{\bar{x}}$ of cylinders (which are prisms on circular bases on x-z plane and height 1/n in the y-direction), one may consider a collection $S_{\bar{x}}$ of right prisms on square bases on x-z plane whose sides are equal to the corresponding deviations (see Fig. 11(a)). In such a case, the task will be to find a right prism R of height 1 and square base (of side to be determined) that has the same volume as the total volume of the collection $S_{\bar{x}}$. When we have found such a right prism R, the side of the square base of R is the population SD σ.

Other generalizations of the collection of prisms $C_{\bar{x}}$ or $S_{\bar{x}}$ are possible. The prisms may have a differently shaped base such as circle-wedge base (see Fig. 11(b)), or right-triangular base (see Fig. 11(c)), or even arbitrary triangular base (see Fig. 11(d)) on x-z plane with sides proportional to the deviations, and all prisms have the same height 1/n. Whatever the chosen shape of the proportional bases, the task is to find a single prism with a similar (same shape, different size) base and total height 1 that has the same volume as the total volume of the collection. When we have found such a prism, the side of the common base (along the x-axis) is the population SD σ.

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3.2.4 The Sample Standard Deviation as the Square Root of Twice the Mean Square of all Pairwise Half-Deviations

When the given *n* numbers are considered as a sample of size *n* (and not as an entire population), the interpretation of the mean $\bar{x}$, the median *m* and the mean deviation remain exactly the same as given in earlier subsections. However, in order to interpret the **sample** SD s_{n-1} we proceed quite differently. In this alternative approach, it becomes clear why the denominator in the definition of sample variance is (*n*–1).

Recall, from Paragraph 1 of Section 1, that the deviation of each of two numbers *a* and *b* from their average is half the deviation between them, or $|a - b|/2$. We shall generalize this concept of a half deviation between two numbers to the case when $n > 2$ numbers are given in order interpret the sample SD s_{n-1}.

Suppose that the given numbers are $x_1, x_2, \ldots, x_n$. We consider all $\binom{n}{2}$ pairs of numbers, and calculate all pairwise half deviations (PHD) given by $|x_i - x_j|/2$, for $1 \leq i < j \leq n$. Next, we combine these (not necessarily distinct) PHD's into one single measure, by computing the mean square (MS) of these PHD's, to get precisely $s_{n-1}^2/2$. We will prove this result in Proposition 4 below using some algebraic manipulations. Having obtained $s_{n-1}^2/2$ in this manner, we will then "reverse the steps" (that is, first double and then take square root) to obtain the sample SD s_{n-1}.

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Proposition 4 The mean square of all $\binom{n}{2}$ pairwise half deviations equals half the sample variance, or $\sum\sum_{1\le i<j\le n} \left(\frac{|x_i - x_j|}{2}\right)^2 \Big/ \binom{n}{2} = \frac{s_{n-1}^2}{2}$.

Proof: The numerator of the left-hand side is the sum of squares of all $\binom{n}{2}$ PHD's. We can interchange indices *i* and *j* to obtain another equivalent sum, and then add in *n* extra terms of the form $|x_i - x_i|^2/2$, for *i*=1, 2, …, *n*, which are, of course, zeroes. Hence,

$$\sum\sum_{1\le i<j\le n} \left(\frac{|x_i - x_j|}{2}\right)^2 = \frac{1}{2}\cdot\frac{1}{4}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2$$

$$= \frac{1}{8}\left\{n\sum_{i=1}^{n}x_i^2 - 2\sum_{i=1}^{n}x_i\cdot\sum_{j=1}^{n}x_j + n\sum_{j=1}^{n}x_j^2\right\}$$

$$= \frac{1}{8}\left\{2n\sum_{i=1}^{n}x_i^2 - 2n^2\bar{x}^2\right\} = \frac{n}{4}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Dividing the extreme expressions above by $\binom{n}{2}$, the number of terms added on the left hand side, we complete the proof. □

The above equivalent definition of sample SD—as **square root of twice the MS of all PHD's** lends itself to a geometric interpretation that we describe in Example 5 below.

Example 5 Suppose that we have a random sample of five values 2, 4, 7, 9 and 13 obtained from some underlying population. We calculate all 10 pairwise deviations, and depict them in Fig. 12(a) as stacked rectangles of width given by the pairwise deviations and height 1/10. Thus, the total height of all rectangles is 1. In Fig. 12(b) we align the width-bisector-lines of all these

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rectangles. In Fig. 12(c), we separate the left- and right-halves of these rectangles, and rotate these collections of rectangles about the width-bisector-line, to form two equivalent collections D_1 and D_2 of right circular cylinders, each having variable cross-sectional radii and height 1 shown in Fig. 12(d). Next, in Fig. 12(e), we consider a new right circular cylinder of height 1 having a volume equal to the total volume of D_1 and D_2 (or twice the volume of each). The radius of the new cylinder is the sample SD s_{n-1}.

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4. Discussion

We have given geometric meanings to familiar statistical measures such as mean, median, MD and SD for a set of n given values. We recalled the existing interpretation of the mean as a fulcrum or center of gravity. Then we reinterpreted the mean as the location of a vertical line superimposed on the graph of the ECDF to equalize the areas of two regions. This latter approach is more favorable to yielding a better guess for the mean than the fulcrum.

Our geometric method of finding the median from the ECDF is nothing new. But, in Fig. 8 we have given an alternative visualization of $\mathrm{MD}(\ell)$ as $A_\ell + B_\ell$. Consequently, we have found a new geometric proof of the well-known optimality property that $\mathrm{MD}(\ell)$ is minimized when $\ell = m$. We must point out that in order to find the exact relation between $\mathrm{MD}(\ell)$ and $\mathrm{MD}(m)$, it is not enough to know only $|\ell - m|$; we must also know how many of the given numbers are between m and ℓ. On the contrary, as shown in (9), population $\mathrm{SD}(\ell)$ is related to population $\mathrm{SD}(\bar{x})$ in a

Pythagorean way: $\mathrm{SD}^2(\ell) = \mathrm{SD}^2(\bar{x}) + (\ell - \bar{x})^2$. Hence, $\mathrm{SD}(\ell)$ is minimized at $\ell = \bar{x}$. This is the underlying reason for the popularity of SD as a measure of spread.

We interpret the population SD geometrically as the radius of a right circular cylinder having a volume equal to that of a solid of revolution generated by revolving the ECDF about $x = \bar{x}$.

Let us return to the central question raised in the Introduction: While $\mathrm{MD}(\bar{x})$ is "the average distance of values from the mean," what exactly is the sample SD? The correct interpretation of the sample SD is the "root *mean* (involving division by (n –1)) squared deviation from the mean;" or, R*M*S$\mathrm{D}(\bar{x})$. Admittedly, it is a rather confusing phrase with two uses of 'mean' having two different meanings. We hope our interpretation of the sample SD as the "root of twice the mean (involving division by $n(n\text{-}1)/2$) square of all pairwise half deviations" will avoid that confusion.

Our geometric interpretations of the mean, the median, the MD, and the population SD for a set of n given numbers also extend to the case of a random variable (either discrete or continuous). Also, our geometric interpretations can be utilized to construct (1) the mean and the SD of a combined sample obtained by aggregating two or more sets of numbers, (2) the mean and the SD of a set of numbers with different frequencies, and (3) the mean and the SD of a histogram. These extensions will be reported in forthcoming papers.

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Figure 1. The mean of the data in Example 1 is shown as a fulcrum at 7

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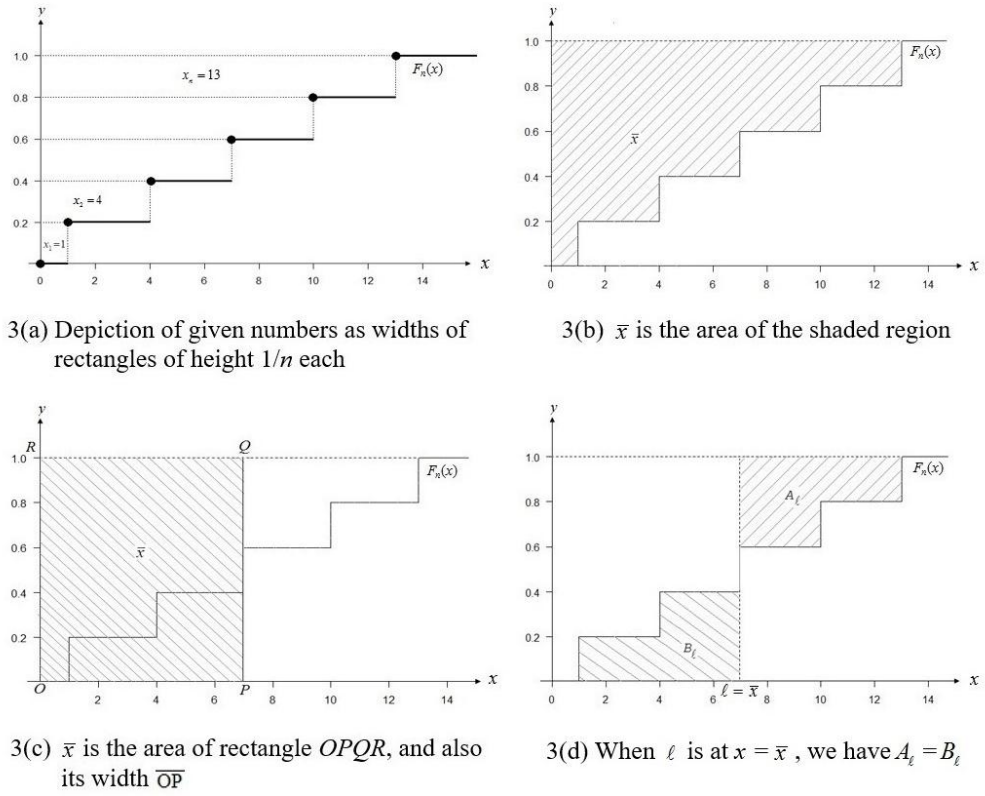


Figure 2. The mean of the modified dataset shown as a fulcrum at 9

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3(a) Depiction of given numbers as widths of rectangles of height $1/n$ each

3(b) $\bar{x}$ is the area of the shaded region

3(c) $\bar{x}$ is the area of rectangle $OPQR$, and also its width $\overline{OP}$

3(d) When ℓ is at $x = \bar{x}$, we have $A_\ell = B_\ell$

Figure 3. An ECDF is utilized (a) to depict the given numbers as widths of rectangles of height $1/n$ each, and (b) to depict $\bar{x}$ as the total area of these rectangles, or as the area of the shaded region. Finally, $\bar{x}$ is depicted as the location of (c) a vertical line PQ, and (d) a vertical line ℓ that equalizes the shaded areas on its two sides

3(a) Depiction of given numbers as widths of rectangles of height $1/n$ each

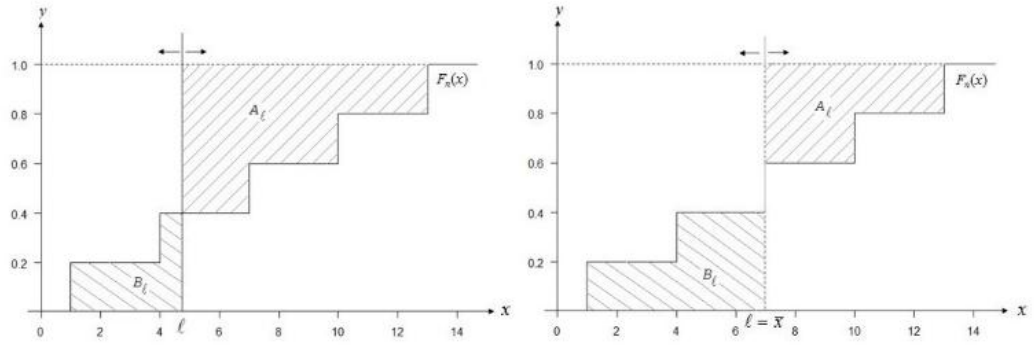
3(b) $\bar{x}$ is the area of the shaded region

3(c) $\bar{x}$ is the area of rectangle $OPQR$, and also its width $\overline{OP}$

3(d) When ℓ is at $x = \bar{x}$, we have $A_\ell = B_\ell$

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4(a) Defing A_ℓ and B_ℓ for an arbitrary ℓ

4(b) Choosing ℓ at $x = \bar{x}$ causes $A_\ell = B_\ell$

Figure 4. The mean is the value $x = \bar{x}$ where the movable vertical line ℓ must be stopped so that regions A_ℓ and B_ℓ will have equal areas

4(a) Defining A_ℓ and B_ℓ for an arbitrary ℓ

4(b) Choosing ℓ at $x = \bar{x}$ causes $A_\ell = B_\ell$

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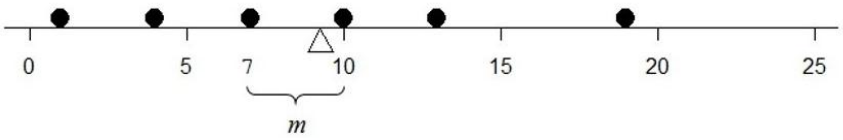
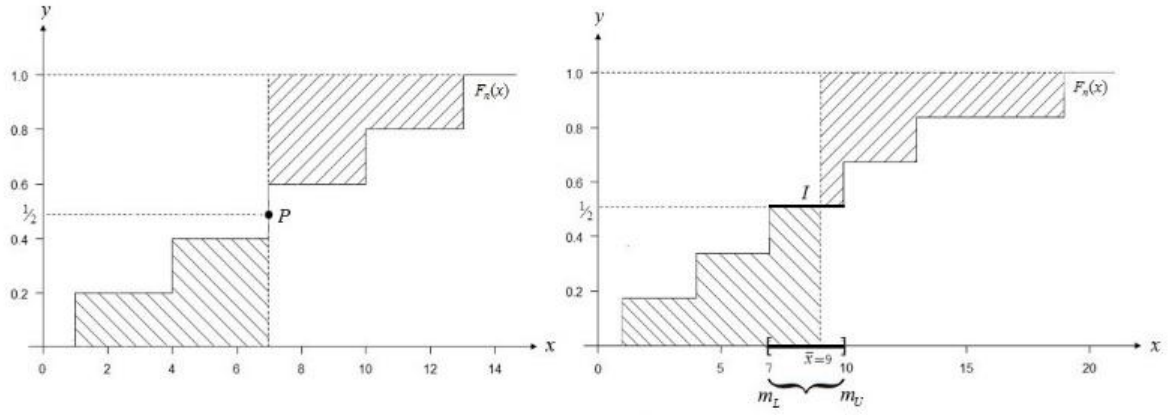


Figure 5. The mean is shown as a fulcrum at 9, and any value in [7, 10] is a median (*m*)

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6(a) $\bar{x} = m = 7$

6(b) $\bar{x} = 9$, m is any value in [7, 10]

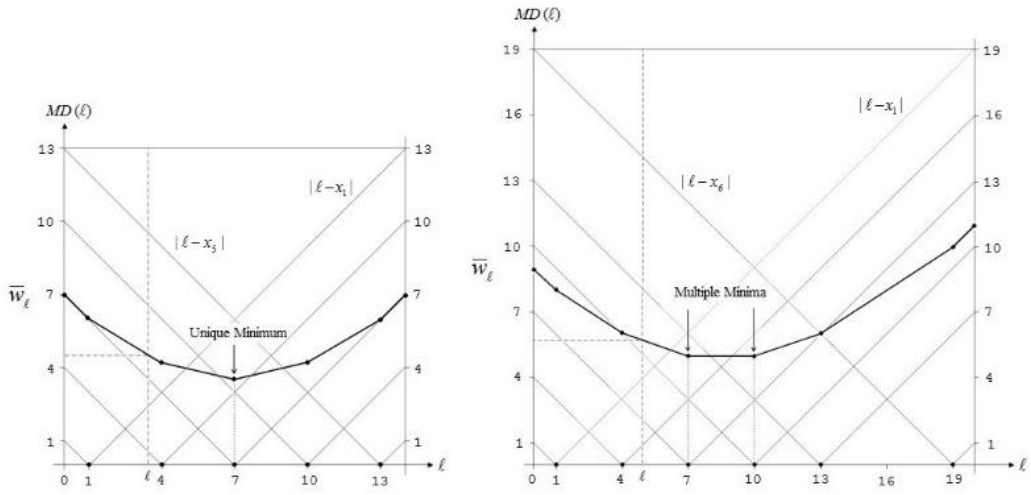
Figure 6. The median is the value $x = m$ where the ECDF intersects the line y=½. The median can be (a) a singleton as in Example 1, or (b) an interval as in Example 3

6(a) $\bar{x} = m = 7$

6(b) $\bar{x} = 9$, m is any value in [7, 10]

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7(a). $MD(\ell) = \overline{w}_\ell$ for data in Example 1

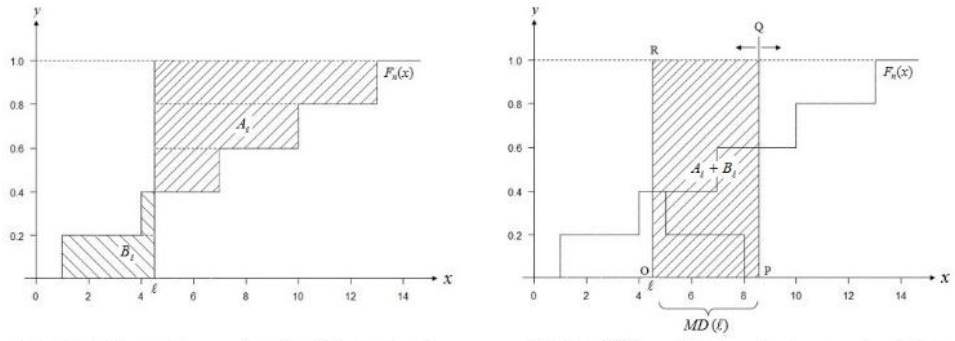
7(b). $MD(\ell) = \overline{w}_\ell$ for data in Example 3

Figure 7. $MD(\ell)$, a piecewise linear function of ℓ, is minimized at the median

7(a). $MD(\ell) = \overline{w}_\ell$ for data in Example 1

7(b). $MD(\ell) = \overline{w}_\ell$ for data in Example 3

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8(a) $MD(\ell)$ is total area $A_\ell + B_\ell$ of shaded regions

8(b) $MD(\ell)$ is width (area) of rectangle OPQR

Figure 8. (a) $MD(\ell) = A_\ell + B_\ell$, and (b) $MD(\ell)$ is the distance between ℓ and PQ

8(a) $MD(\ell)$ is total area $A_\ell + B_\ell$ of shaded regions

8(b) $MD(\ell)$ is width (area) of rectangle OPQR

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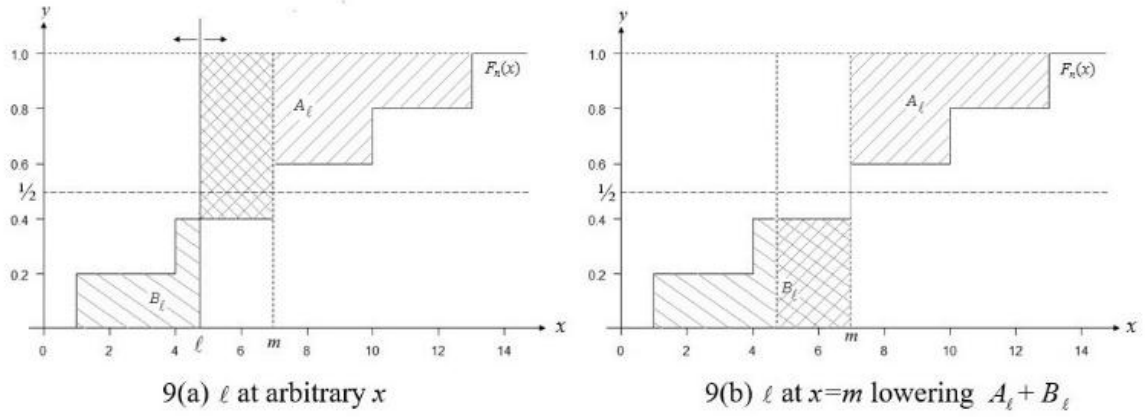


Figure 9. As the vertical line moves from $\ell < m$ to *m*, the reduction in A_ℓ, the checkered area in Fig. 9(a), exceeds the gain in B_ℓ, the checkered area in Fig. 9(b). Therefore, $MD(\ell) = A_\ell + B_\ell$ is minimized when ℓ=*m*

9(a) ℓ at arbitrary *x*

9(b) ℓ at *x*=*m* lowering $A_\ell + B_\ell$

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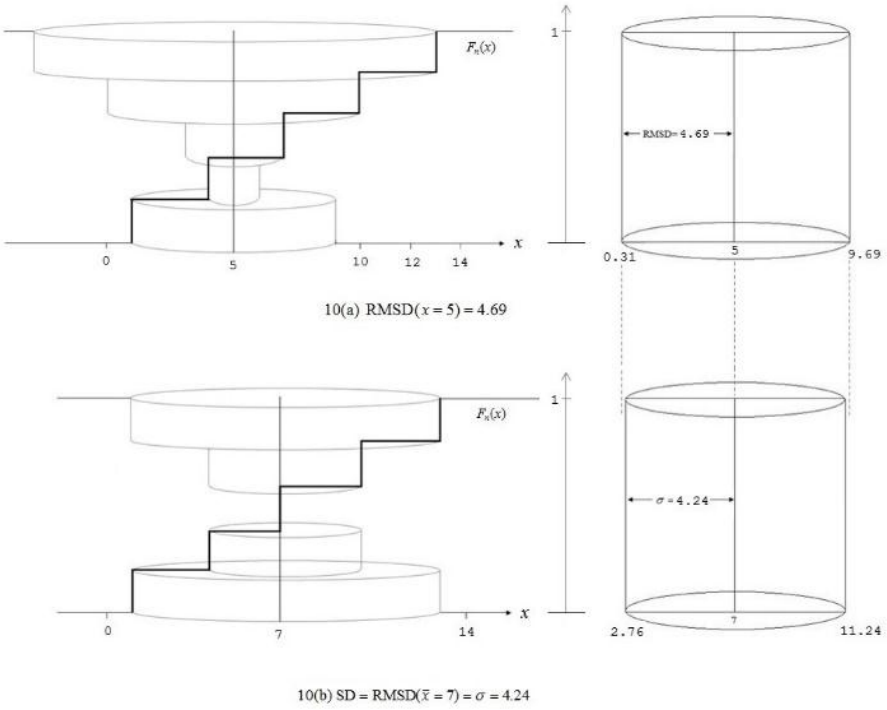


Figure 10. The RMSD ($x = \ell$) is the radius of a right circular cylinder of height 1 whose volume equals that of the solid of revolution when the graph of the ECDF is rotated about the vertical line through an arbitrary $x = \ell$. The volume (hence, radius) of the resultant cylinder in Fig. 10(a) where *x* is arbitrary, exceeds that in Fig. 10(b) where $x = \bar{x}$.

10(a) RMSD (*x*=5)=4.69

10(b) SD=RMSD ($\bar{x} = 7$) = s = 4.24

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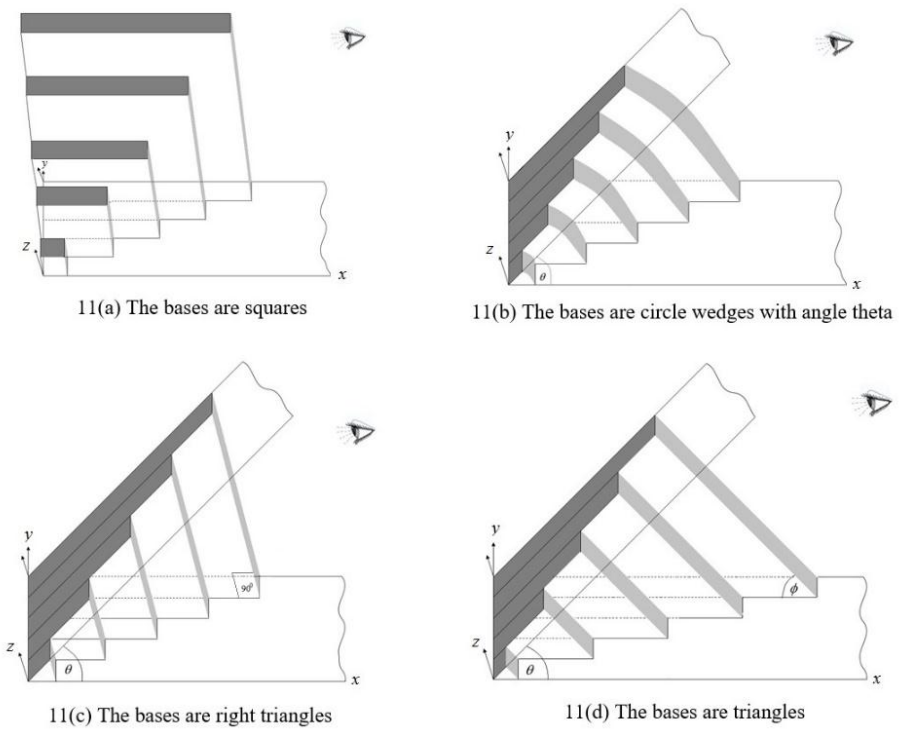


Figure 11. Several different choices of bases (parallel to *x*-*z* plane) are shown here: A collection of prisms each of height 1/*n* and having variable but proportional bases has a total volume that must be attained by a single prism of total height 1 and a similar constant base (parallel to *x*-*z* plane). Then the side of the common base (along the *x*-axis), irrespective of the shape of the base, is the population SD σ

11(a) The bases are squares

11(b) The bases are circle wedges with angle theta

11(c) The bases are right triangles

11(d) The bases are triangles

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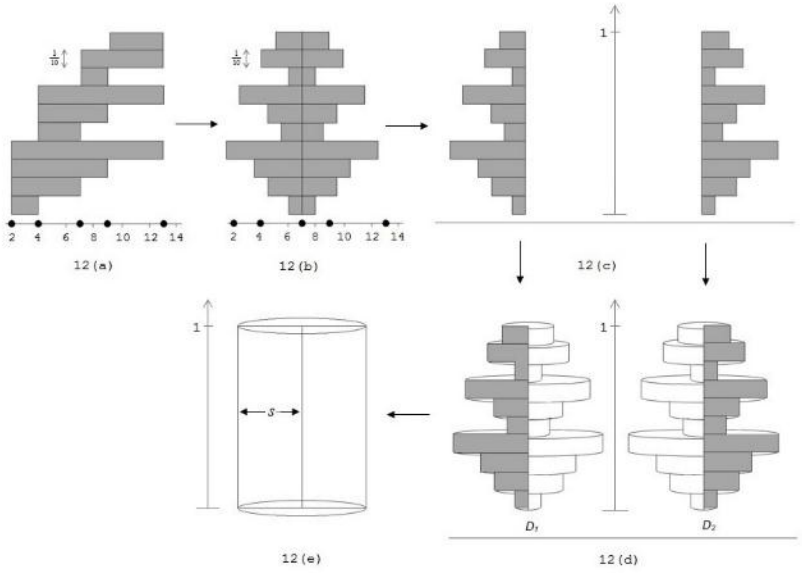


Figure 12. Illustrating the sample standard deviation $s=s_{n-1}$ as the square root of twice the mean square of half deviations for all possible pairs formed from the given numbers in Example 5

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Table 1. Computations of mean deviations in Examples 1 through 3

	Mean $\bar{x}$	**Median *m***	$MD(\mathbf{0})$	$MD(\bar{x})$	$MD(m)$
Example 1	7	7	7	3.6	3.6
Example 2	9	7	9	6.0	5.6
Example 3	9	[7, 10]	9	5.0	5.0

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